

ME436A-IS: Plasma Engineering

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Paschen Curve Graphing Assignment

Instructor: Professor Wright

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1. Introduction

In this assignment, we:

- Plot the Paschen curves for air, argon, and carbon dioxide.
- Plot points representing the breakdown voltage of air at different selected pd values.
- Derive $\min(V_{BD})$.

All relevant Paschen Law constants (A and B) to plot the Paschen curves of relevant gases were derived from Table 1 of [“Suppression of electrical breakdown phenomena in liquid TriMethyl Bismuth based ionization detectors”](#)¹.

Gas	A ($\text{mbar}^{-1} \cdot \text{cm}^{-1}$)	B ($\text{V} \cdot \text{mbar}^{-1} \cdot \text{cm}^{-1}$)
He	2	26
Ar	9	135
Air	11	274
Ne	3	751
CO ₂	15	350

Table 1: *Table of Paschen Constants, derived from “Suppression of electrical breakdown phenomena in liquid TriMethyl Bismuth based ionization detectors”*

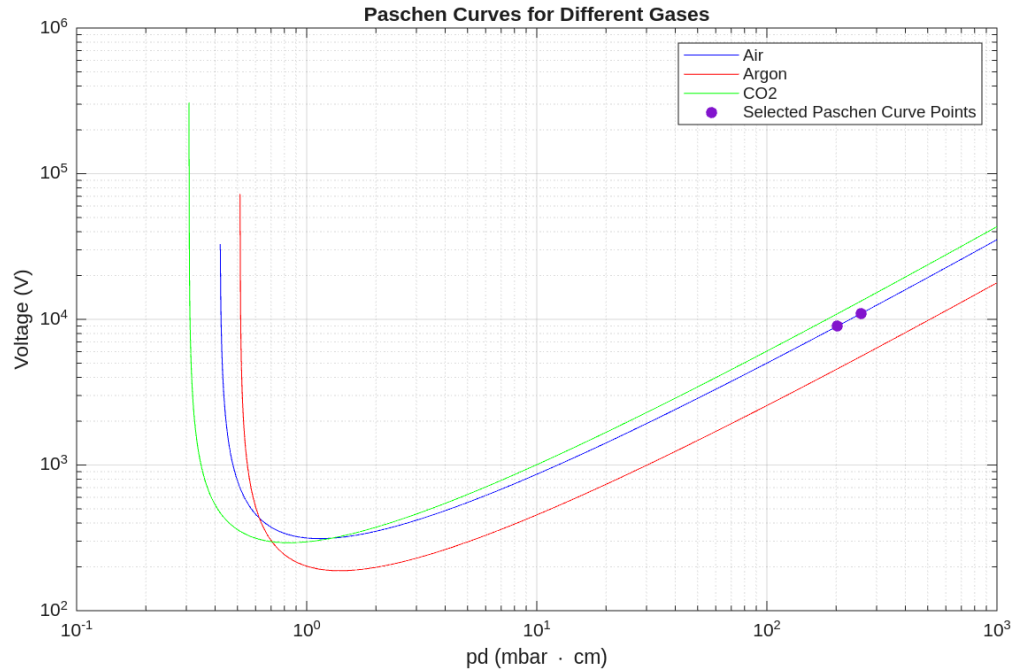


Figure 1: *Paschen curves for air, argon, and carbon dioxide, with selected air breakdown-voltage points.*

2. Derivation of $\min(V_{BD})$

We claim:

$$\min(V_{BD}) = \frac{Be \ln(\gamma^{-1} + 1)}{A}$$

Recall the equation for Paschen's Law:

$$V_{BD}(pd) = \frac{Bpd}{\ln(Apd) - \ln(\ln(\gamma^{-1} + 1))} \quad (1)$$

To find $\min(V_{BD})$, take the derivative of V_{BD} with respect to pd :

$$\begin{aligned} \frac{dV_{BD}}{d(pd)} &= \frac{B(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1))) - Bpd(pd)^{-1}}{(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2} \\ \frac{dV_{BD}}{d(pd)} &= \frac{B(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1)}{(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2} \end{aligned} \quad (2)$$

Recall that V_{BD} has a local extrema at $(pd = x)$ if:

$$\left. \frac{dV_{BD}}{d(pd)} \right|_{pd=x} = 0$$

Note that

$$\begin{aligned} &(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1 \neq \ln(Apd) - \ln(\ln(\gamma^{-1} + 1))) \\ \Rightarrow &((\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1 = 0) \Rightarrow (\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) \neq 0)) \end{aligned}$$

Therefore, we can compute the zero(es) of the derivative of V_{BD} with respect to pd by calculating the zero(es) of $(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1)$. Let:

$$0 = \ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1$$

$$\Rightarrow 1 = \ln\left(\frac{Apd}{\ln(\gamma^{-1} + 1)}\right)$$

$$\Rightarrow e = \frac{Apd}{\ln(\gamma^{-1} + 1)}$$

$\Rightarrow pd = \frac{e \ln(\gamma^{-1} + 1)}{A} \quad (3)$

Therefore, $\left(x = \frac{e \ln(\gamma^{-1} + 1)}{A}\right)$ is a local extrema of V_{BD} .

We also claim that $(V_{BD}(x) = \min(V_{BD}))$, which we can prove rigorously by analyzing the second derivative of V_{BD} with respect to pd .

Now, consider the second derivative of V_{BD} with respect to pd (done with Wolfram Alpha):

$$\frac{d^2 V_{BD}}{d(pd)^2} = \frac{B}{pd(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2}$$

In the context of a Paschen Curve:

$$((p > 0) \wedge (d > 0)) \Rightarrow (pd > 0)$$

Additionally, per Table 1, $(B > 0)$. Since $\left(\left(pd(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2\right) > 0\right)$, this then implies that $\left(\frac{d^2 V_{BD}}{d(pd)^2} > 0\right)$. Therefore, since the graph of V_{BD} is concave up, then the extrema $\left(x = \frac{e \ln(\gamma^{-1} + 1)}{A}\right)$ is an absolute minimum.

Now, compute V_{BD} at the local extrema present in Equation 3:

$$V_{BD}(x) = \frac{B\left(\frac{e \ln(\gamma^{-1} + 1)}{A}\right)}{\ln\left(A\left(\frac{e \ln(\gamma^{-1} + 1)}{A}\right)\right) - \ln(\ln(\gamma^{-1} + 1))}$$

$$V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A(\ln(e \ln(\gamma^{-1} + 1)) - \ln(\ln(\gamma^{-1} + 1)))}$$

$$V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A(1 + \ln(\ln(\gamma^{-1} + 1)) - \ln(\ln(\gamma^{-1} + 1)))}$$

$\min(V_{BD}) = V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A} \tag{4}$
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3. Appendix MATLAB Code

Here was the following code to generate the MATLAB figure:

```
%% Params. for Paschen Curve

pd = logspace(-1,3,2000); % 0.1 to 1000 on a log-spaced grid
gamma = 0.01; %dimensionless; assuming electrodes are made out of stainless
steel

Patm = 1013.25 %mbar
x = [Patm*0.2, (Patm+(4*68.948))*0.2] %mbar*cm

% Air
A_Air = 11; %m*bar^(-1)*cm^(-1)
B_Air = 274; %V*bar^(-1)*cm^(-1)

% Calculate the Paschen curve for Air
V_Air = B_Air .* pd ./ (log(pd) + log(A_Air) - log(log((1 / gamma) + 1)));
V_Air((log(pd) + log(A_Air) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% Ar
A_Ar = 9; %m*bar^(-1)*cm^(-1)
B_Ar = 135; %V*bar^(-1)*cm^(-1)
V_Ar = B_Ar .* pd ./ (log(pd) + log(A_Ar) - log(log((1 / gamma) + 1)));
V_Ar((log(pd) + log(A_Ar) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% CO2
A_CO2 = 15 %m*bar^(-1)*cm^(-1)
B_CO2 = 350 %V*bar^(-1)*cm^(-1)
V_CO2 = B_CO2 .* pd ./ (log(pd) + log(A_CO2) - log(log((1 / gamma) + 1)));
V_CO2((log(pd) + log(A_CO2) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% Plot the Paschen curves for Air, Argon, and CO2
figure;
loglog(pd, V_Air, 'b', 'DisplayName', 'Air');
hold on;
```

```
loglog(pd, V_Ar, 'r', 'DisplayName', 'Argon');
loglog(pd, V_CO2, 'g', 'DisplayName', 'CO2');
y_air_pts = interp1(pd, V_Air, x, 'makima');
scatter(x, y_air_pts, 'filled', 'DisplayName', 'Selected Paschen Curve Points');
hold off;
grid on;
xlabel('pd (mbar \cdot cm)');
ylabel('Voltage (V)');
title('Paschen Curves for Different Gases');
legend('show');
```

4. Appendix Typst Code

If you're interested in seeing the code I used to produce this document, here's my template .typ file:

```
#show figure: set block(below: (2.0em))

#let numbered_eq(content) = math.equation(
  block: true,
  numbering: "(1)",
  content,
)

#let date = datetime.today()

#show table.cell.where(y: 0): set text(weight: "bold")

#let conf(
  course_code: [XYZ 123],
  course_title: [Course Title],
  course_color: gradient.linear(..color.map.plasma),
  authors: ("Allen Rakhimov", "Additional Authors"),

  title: [Document Title],
  instructors: ("Professor 1", "Professor 2",),
  month: [datetime.now().month.repr:long],
  year: [2025],
  doc,
) = {
  set page(
    paper: "us-letter",
    margin: (1in),
  )

  set text(
    font: "New Computer Modern",
    size: 12pt
```

```
)

set par(
  leading: 1.30em
)

show link: url => underline(text(fill: rgb(0, 0, 255))[#url])

let date = [#month #year]
let left_head = [#course_code: #title]
let course = [#course_code: #course_title]

// Dynamic right header :)
let right_head = [
  Joshua Davidov \
]

// Course Title
text(course_color, size: 24pt)[
  #box(course)
]

v(1em, weak: true)

for author in authors [
  #if author != authors.at(-1) [
    #author
  ] else [
    #author
  ]
]

// Document Title
place(
```



```
horizon + center,
text(size: 24pt)[#title]
)

place(
bottom + right,
[
  #if instructors.len() > 1 [
    Instructors:
  ] else [
    Instructor:
  ]
  #for instructor in instructors [
    #if instructor != instructors.at(-1) [
      #instructor\
    ] else [
      #instructor
    ]
  ]
]

#date
]
)

pagebreak()
// Create header but only after title page
counter(page).update(1)
set page(
  numbering: "1",
  header: [
    #set text(10pt)
    #left_head
    #h(1fr) Davidov, J. 2026
  ]
)
```

```
#v(0.6em, weak: true)
#line(length: 100%, stroke: 0.5pt)
],
)
doc
}
```

5. References

- [1] B. Gerke et al., “Suppression of electrical breakdown phenomena in liquid TriMethyl Bismuth based ionization detectors,” *Journal of Instrumentation*, vol. 17, no. 09, p. P09029, Sep. 2022, doi: 10.1088/1748-0221/17/09/P09029.